

Tracing the Roots of Environmental Racism in Houston's Fifth Ward

Carlos Mendieta | Pharis Fellowship

Background

- In 2019 the Texas Department of Health declared the Fifth Ward/Kashmere Gardens, (a historically black community in Houston), as a cancer cluster.
- A cancer cluster is the occurrence of a greater than expected number of cancer cases among a group of people in a specific geographic area and time period.
- Residents of the Fifth Ward/Kashmere Gardens neighborhood in Houston have raised concerns of living in close proximity to industrial sites, releasing environmentally hazardous and carcinogenic chemicals into their community, namely the Union Pacific Railroad System. According to City of Houston Mayor Sylvester Turner, childhood leukemia cases are 350 percent higher than expected in this region.

Methodology

Hypothesis: Industrial sites in Houston have been disproportionately placed in less affluent areas of the city. Far proximity from a concentration of industrial sites can serve as an indicator of higher housing value, over time pushing low-income minorities closer to environmentally toxic areas, increasing the likelihood of being exposed to carcinogens.

Data: Filtered and merged data from the Harris County Appraisal District for single family residential market values from the years 2005 and 2022. Filtered the “Toxics Release Inventory” from the Environmental Protection Agency that answered “YES” to Carcinogenic status and “YES” to PBT status (persistent, bioaccumulative, and toxic chemicals).

Visualization: Made a map to see residential growth from 2005-2022. To include an aspect of socioeconomic status per census tract, created a formula for a map gradient. Formula = Median Tract change in Housing Market Value / Median Tract Family Income.

Limitations

- Harris County Appraisal District data only runs from 2005-2023.
- EPA Toxics Release Inventory only contains reported industrial sites, potential for many unreported sites in Houston
- Median Family Income calculation at times may have too much influence on mapped census tracts’ gradients.

Results

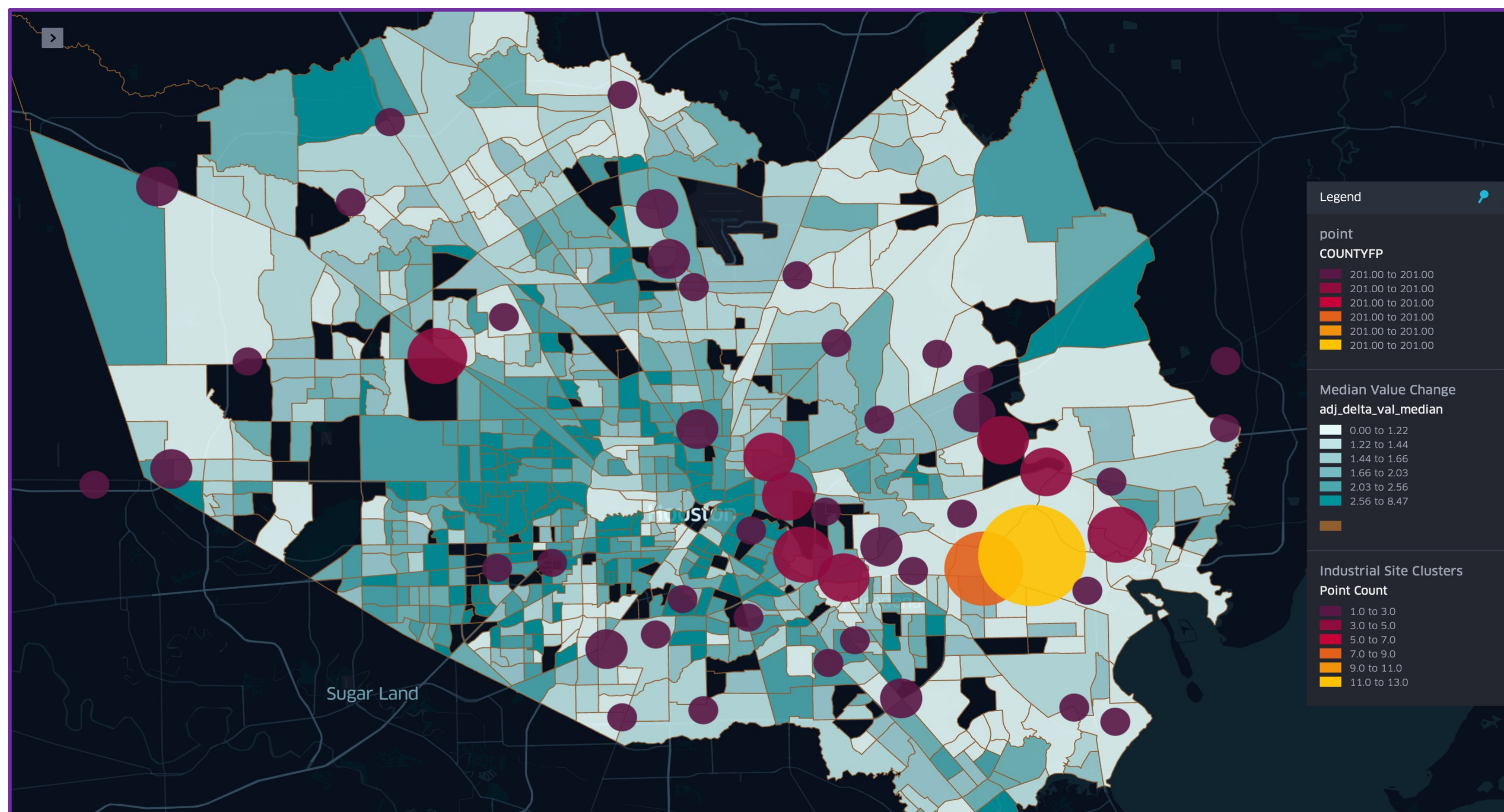


Figure 1: Map of median home value change from 2005-2022 and carcinogenic industrial sites per census tract in Harris County.

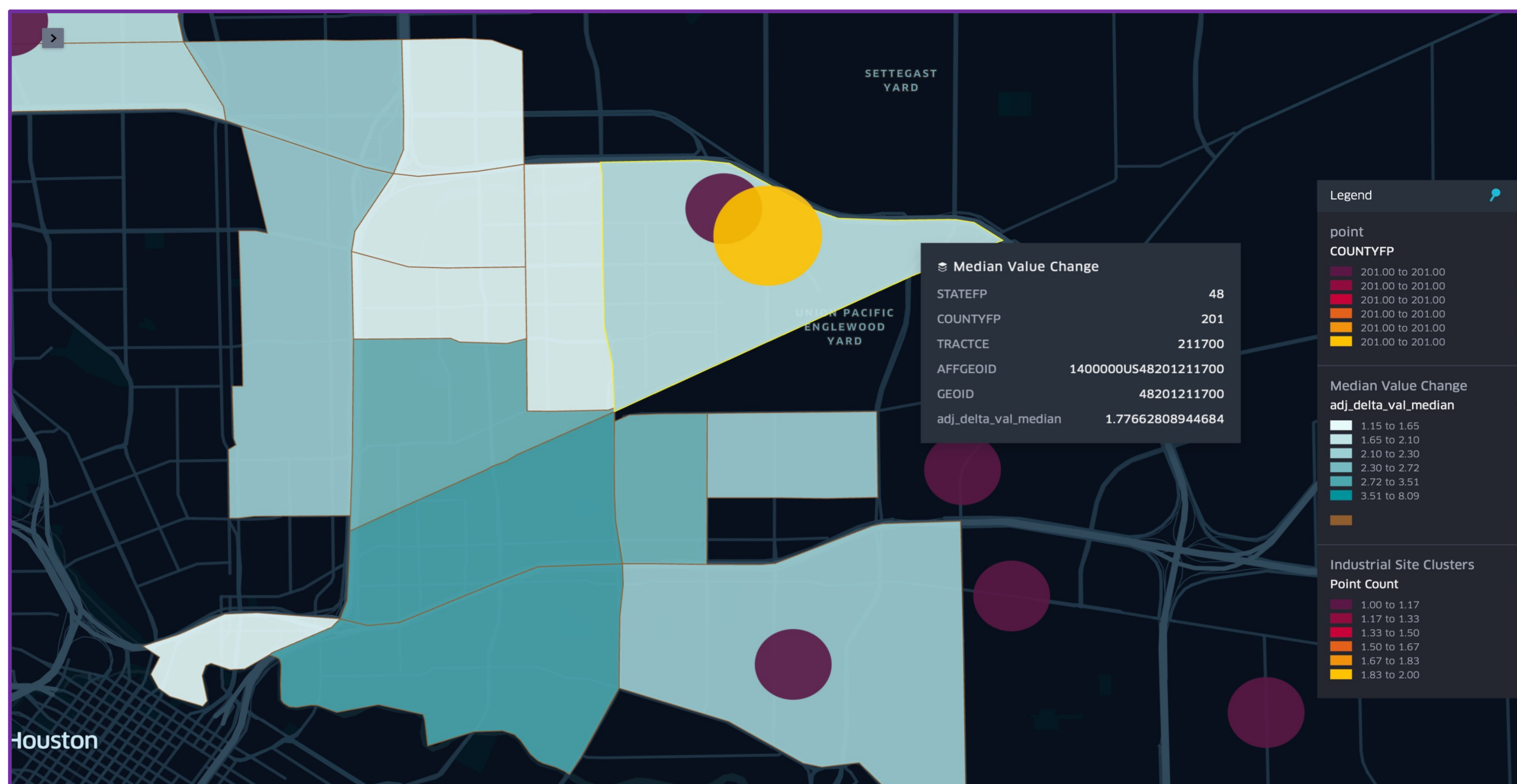


Figure 2: Figure 1 zoomed into Fifth Ward/Kashmere Gardens census tracts.

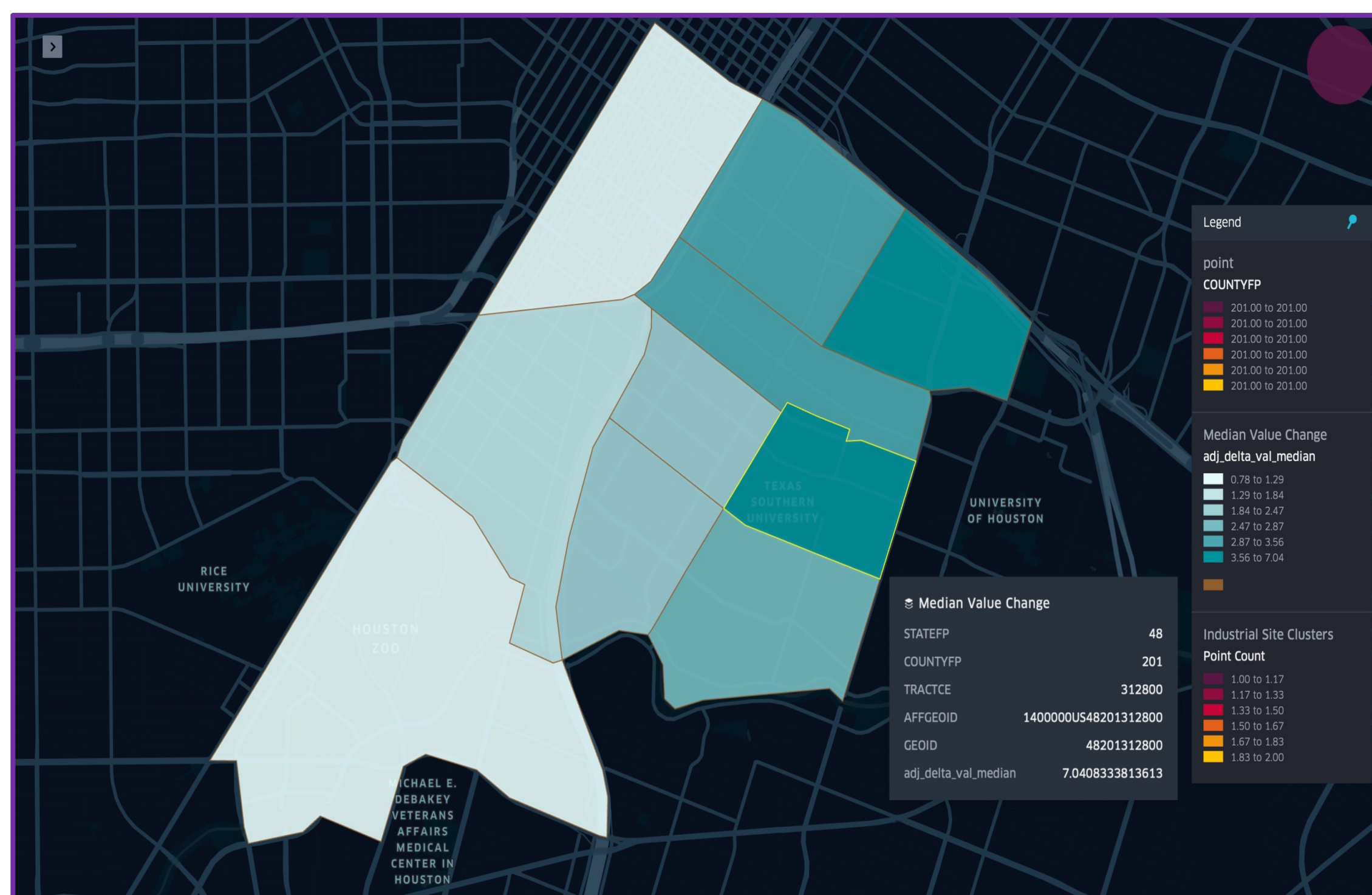


Figure 3: Figure 1 zoomed into Third Ward census tracts.

How to Read Map:

- Different shades of teal census tracts correspond to the values listed on the legend under “Median Value Change”. This number is the product received from the formula outlined under “Methodology – Visualization”. This number serves as a factor to see both how much a property’s value grew from 2005 – 2022, and how much more the average property per census tract costs in relativeness to the median family income in that census tract.
- For example, in **Figure 3**, the Third Ward census tract outlined in yellow has an “adj_delta_val_median” value of 7.04... What this means is from 2005 – 2022, the median value of a single-family residential home has increased over 7 times the median family income in that census tract.
- The dots on the map correspond to “Industrial Site Clusters” in the legend. These represent the industrial sites outlined in “Methodology – Data”. The number means the amount of those industrial sites inside of that dot.

Conclusions

- Generally, there is a correlation between the presence of industrial sites and lower residential property growth in Houston. Census tracts with at least one industrial site were generally shown to have a lower increase in single-family residential housing market values from 2005-2022, notably prevalent in Houston’s east end.
- Compared to other affluent and non-affluent neighborhoods throughout Houston, Fifth Ward/Kashmere Gardens showed considerably less growth in housing market values from 2005-2022. Seemingly, these areas also contain many industrial sites.
- In the Fifth Ward/Kashmere Gardens, housing values closest to the Union Pacific Railyard have shown lower growth rates compared to other census tracts within the same neighborhood, with the highest growing properties being furthest away from the railyard and other industrial sites.
- The data visualized does provide support to my hypothesis that individuals in Houston that could only afford the cheapest housing accommodations were potentially moved into generally more environmentally toxic areas, posing as cancer and other fatal health risks. However, these findings cannot serve as a definitive causation as many other factors could also play a role.

Future Directions

- Aggregating data to be more specific to population demographics, utilizing census blocks instead of tracts.
- Exploring how healthcare access varies across specific neighborhoods. Does redlining infrastructure affect whether areas have more or less access?
- View issues from resident perspective – what historical influences do they believe has caused this health crisis?

Acknowledgements

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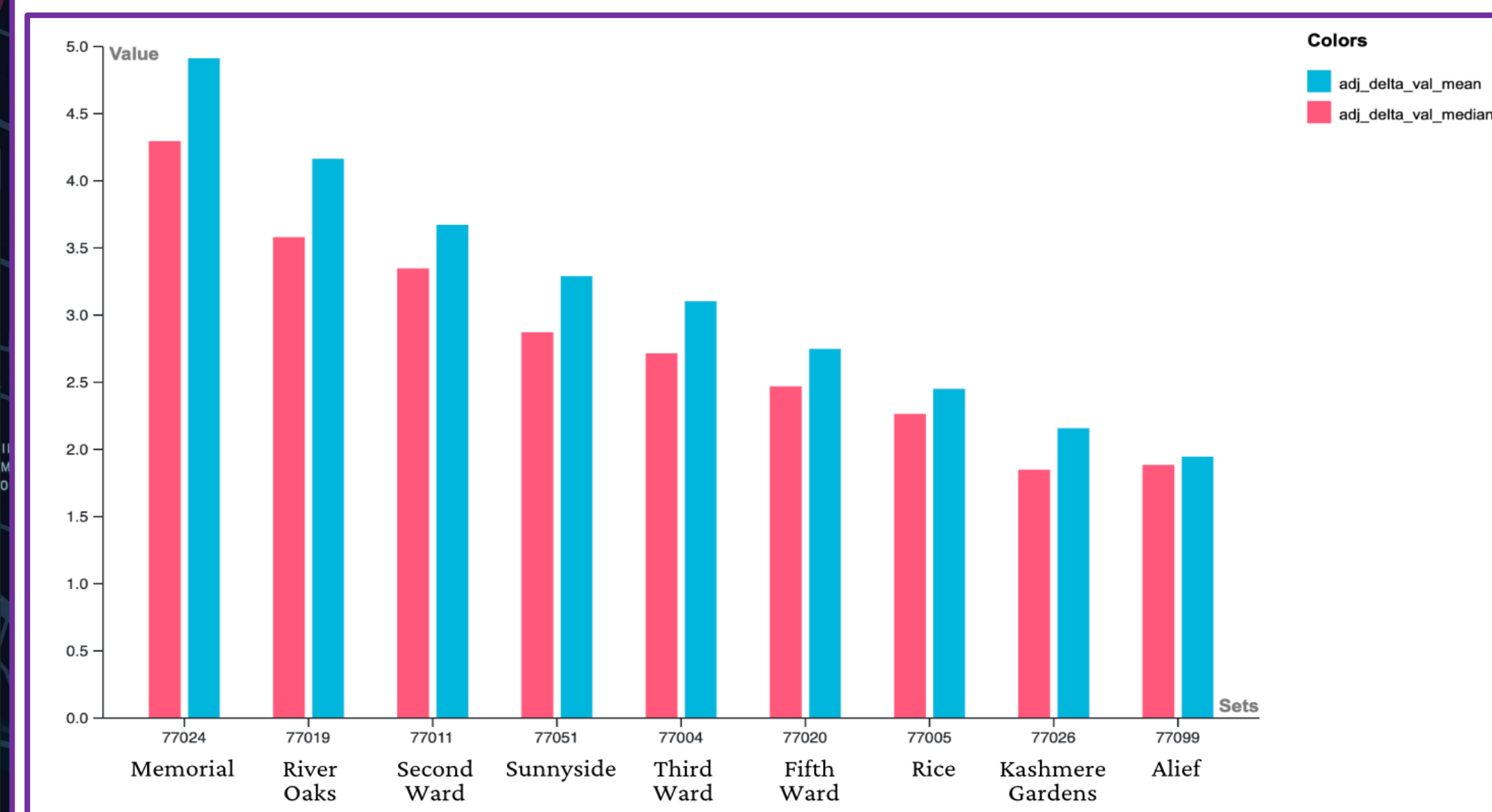


Figure 4: Mean & Median home value change from 2005-2022.



References & Contact